STA Sequence

Bx/mmCIF Format
Bx/mmCIF Format (gz)
aryCIF Format (gz)

gacy PDB Format
gacy PDB Format (gz)

BML/XML Format (gz)

ucture Factors (CIF)
ucture Factors (CIF - gz)

lidation Full (PDF - gz)
lidation (XML - gz)
lidation (CIF - gz)
lidation 2fo-fc coefficients (CIF - gz)
lidation fo-fc coefficients (CIF - gz)

ological Assembly 1 (CIF - gz)
ological Assembly 1 (PDB - gz)

0002vgb

RUVATE KINASE

2VGB/pdb

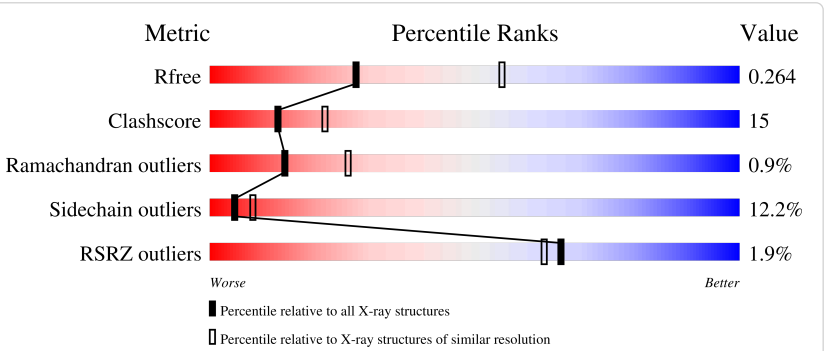
BL21

07-11-20

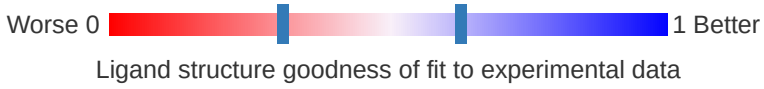
hiarelli, L., Fortin, R., Dolzan, M., Galizzi, A., Abraham, D.J., Wang, C., Bianchi, P., Zanella, A., Mattevi, A.

Starting Model: experimental
View more details

wwPDB Validation 3D Report | Full Report



Ligand Structure Quality Assessment



Ligand structure goodness of fit to experimental data

This is version 2.0 of the entry. See complete history.

Literature

Download Primary Citation

Structure and Function of Human Erythrocyte Pyruvate Kinase. Molecular Basis of Nonspherocytic Hemolytic Anemia.

Valentini, G., Chiarelli, L.R., Fortin, R., Dolzan, M., Galizzi, A., Abraham, D.J., Wang, C., Bianchi, P., Zanella, A., Mattevi, A.
(2002) J Biol Chem **277**: 23807

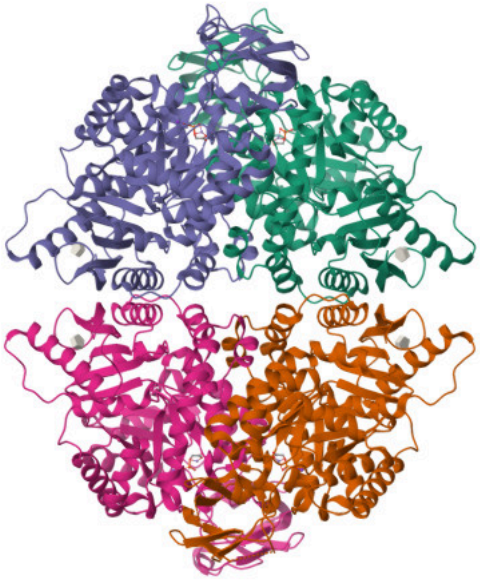
PubMed: 11960989 Search on PubMed
DOI: https://doi.org/10.1074/jbc.M202107200
Primary Citation Related Structures:
2VGB, 2VGF, 2VGG, 2VGI

PubMed Abstract:
Deficiency of human erythrocyte isozyme (RPK) is, together with glucose-6-phosphate dehydrogenase deficiency, the most common cause of the nonspherocytic hemolytic anemia. To provide a molecular framework to the disease, we have solved the 2.7 A resolution crystal structure of human RPK in complex with fructose 1,6-bisphosphate, the allosteric activator, and phosphoglycolate, a substrate analogue, an...
View More

Organizational Affiliation:

Dipartimento di Genetica e Microbiologia, Università di Pavia, via Abbiategrasso 207, 27100 Pavia, Italy. giovale@unipv.it

Biological Assembly 1



Explore in 3D: Structure | Sequence Annotations | Electron Density | Validation Report | Ligand Interaction (FBP)

Global Symmetry: Dihedral - D2 (Explore in 3D)
Global Stoichiometry: Homo 4-mer - A4

Find Similar Assemblies

Biological assembly 1 assigned by authors and generated by PQS (software)

Macromolecule Content

- Total Structure Weight: 230.26 kDa
- Atom Count: 15,612
- Modeled Residue Count: 2,037
- Deposited Residue Count: 2,112
- Unique protein chains: 1

Macromolecules

Find similar proteins by: Sequence | 3D Structure

Entity ID: 1

Molecule	Chains	Sequence Length	Organism	Details
PYRUVATE KINASE ISOZYMES R/L	A, B, C, D	528	Homo sapiens	**Mutation(s)**: 0 **EC:** 2.7.1.40

UniProt & NIH Common Fund Data Resources

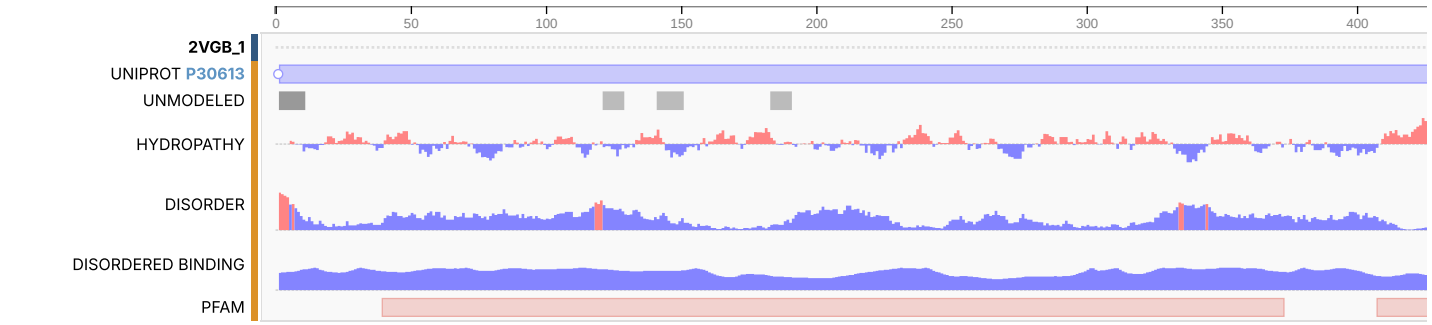
Find proteins for P30613 *(Homo sapiens)* Explore P30613

PHAROS: P30613 GTEx: ENSG00000143627

Entity Groups	
Sequence Clusters	30% Identity 50% Identity 70% Identity 90% Identity 95% Identity 100% Identity
UniProt Group	P30613

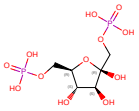
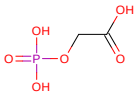
Sequence Annotations

Reference Sequence 2VGB_1



Small Molecules

Ligands 4 Unique

ID	Chains	Name / Formula / InChI Key	2D Diagram	3D Interactions
FBP Query on FBP Download: Ideal Coordinates CCD File Instance Coordinates	E [auth A], I [auth B], M [auth C], Q [auth D]	**1,6-di-O-phosphono-beta-D-fructofuranose** $C_6 H_{14} O_{12} P_2$ RNBGYGVWRKECFJ-ARQDHWQXSA-N		Interactions Interactions & Density
PGA Query on PGA Download: Ideal Coordinates CCD File Instance Coordinates	F [auth A], J [auth B], N [auth C], R [auth D]	**2-PHOSPHOGLYCOLIC ACID** $C_2 H_5 O_6 P$ ASCFNMCAHFUBCO-UHFFFAOYSA-N		Interactions Interactions & Density
MN Query on MN Download: Ideal Coordinates CCD File Instance Coordinates	H [auth A], L [auth B], P [auth C], T [auth D]	**MANGANESE (II) ION** Mn WAEMQWOKJMHJLA-UHFFFAOYSA-N	Mn^{+2}	Interactions Interactions & Density
K Query on K Download: Ideal Coordinates CCD File Instance Coordinates	G [auth A], K [auth B], O [auth C], S [auth D]	**POTASSIUM ION** K NPYPAHLBTDXSSS-UHFFFAOYSA-N	K^{+}	Interactions Interactions & Density

Experimental Data & Validation

Experimental Data

Method: X-RAY DIFFRACTION
Resolution: 2.73 Å
R-Value Free: 0.271 (Depositor), 0.264 (DCC)
R-Value Work: 0.227 (Depositor), 0.221 (DCC)
R-Value Observed: 0.229 (Depositor)

Space Group: P 1 2_1 1

Unit Cell:

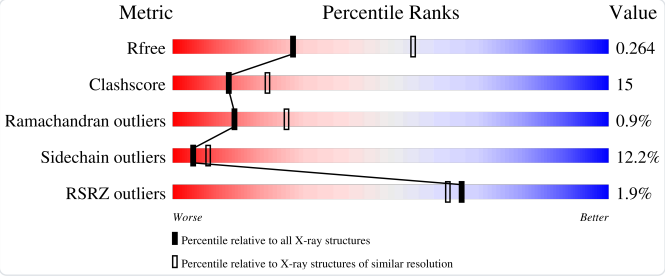
Length (Å)	Angle (°)
a = 74.413	α = 90
b = 172.058	β = 92.46
c = 85.512	γ = 90

Software Package:

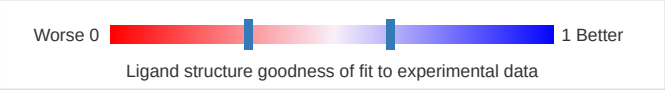
Software Name	Purpose
REFMAC	refinement
MOSFLM	data reduction
CCP4	data scaling
AMoRE	phasing

Structure Validation

View Full Validation Report



Ligand Structure Quality Assessment



View more in-depth experimental data

Entry History

Deposition Data

Released Date: 2007-11-20
Deposition Author(s): Valentini, G., Chiarelli, L., Fortin, R., Dolzan, M., Galizzi, A., Abraham, D.J., Wang, C., Bianchi, P., Zanella, A., Mattevi, A.
This entry supersedes: 1LIU

Revision History (Full details and data files)

- **Version 1.0: 2007-11-20**
  Type: Initial release
- **Version 1.1: 2011-07-13**
  Changes: Advisory, Version format compliance
- **Version 1.2: 2018-05-16**
  Changes: Data collection
- **Version 1.3: 2019-05-08**
  Changes: Data collection, Experimental preparation
- **Version 1.4: 2020-07-29**
  Type: Remediation
  Reason: Carbohydrate remediation

Changes: Data collection, Derived calculations, Other, Structure summary

- **Version 1.5: 2023-12-13**
  Changes: Data collection, Database references, Refinement description, Structure summary
- **Version 2.0: 2024-05-01**
  Changes: Atomic model, Derived calculations

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Nucleic Acid Knowledgebase

RCSB PDB Core Operations are funded by the U.S. National Science Foundation (DBI-2321666), the US Department of Energy (DE-SC0019749), and the National Cancer Institute, National Institute of Allergy and Infectious Diseases, and National Institute of General Medical Sciences of the National Institutes of Health under grant R01GM157729. RCSB PDB uses resources of the National Energy Research Scientific Computing Center (NERSC), a Department of Energy User Facility.